

<u>Steps</u>	<u>Reason</u>
1. $(p \wedge q) \rightarrow r$	Premise
2. $\neg p \rightarrow \neg s$	Premise
3. $\neg q \rightarrow \neg t$	Premise
4. $\neg r$	Premise
5. $e \rightarrow (t \wedge \neg t)$	Premise
6. $\neg(p \wedge q)$	Modus tollens on 4, 1
7. $\neg p \vee \neg q$	De Morgan's law on 6
8. $\neg(\neg p) \vee s$	Conditional disjunction equivalence on 2
9. $p \vee s$	Double-negation law on 8
10. $s \vee \neg q$	Resolution on 9, 7
11. $\neg q \vee s$	Commutative law on 10
12. $\neg(\neg q) \vee t$	Conditional disjunction equivalence on 3
13. $q \vee t$	Double-negation law on 12
14. $t \vee s$	Resolution on 13, 11
15. $s \vee t$	Commutative law on 14
16. $\neg(\neg(s \vee t))$	Double negation law on 15
17. $\neg(\neg s \wedge \neg t)$	De Morgan's law on 16
18. $\neg e$	Modus tollens on 17, 5

Some references for Section 1.1 to 1.6

1. The Mathematical Analysis of Logic — George Boole
2. The Laws of Thought — George Boole
3. Organon — Aristotle
4. April 1984 issue — Annals of the History of Computing
5. Read Symbolic Logic by Lewis Carroll
6. Logic in Computer Science — M. Huth and M. Ryan
7. Introduction to Mathematical Logic — E. Mendelson
8. Sets, Logic and Axiomatic Theories — R. R. Stoll